

Hosting facts (PDPL boundary, regions, domains)

SECTION 06-operations-it TYPE document STATUS **active** OWNER Founder UPDATED 2026-08-19

Single-page reference for where Fly GACA actually runs. Everything else in `06-operations-it/` keys off this page — check it before reaching for the longer planning docs, and before restating a region, a datastore or a repo path anywhere else in the tree.

The stack in one table

Concern	What runs it
SPA (<code>dist/</code>)	Cloud Storage bucket behind an HTTPS load balancer
API (<code>server/</code>)	Cloud Run service — a single Express 5 app, region <code>me-central2</code>
Datastore	Cloud SQL for PostgreSQL, same region
Sessions	HS256 JWT in an HttpOnly cookie, signed by the API (<code>SESSION_SECRET</code>)
Sign-in	Email + password (scrypt) and Google, via server-side OAuth
Payments	Moyasar — mada, Apple Pay, cards (hosted widget + server-to-server confirm)
Renewals	Cloud Scheduler → <code>POST /api/billing/renew</code> , carrying <code>CRON_SECRET</code>
AI (Captain Adel)	Gemini via Genkit, RAG over the GACAR corpus
Transactional email	Any Resend-compatible endpoint (<code>MAIL_ENDPOINT</code> / <code>MAIL_API_KEY</code>)
Secrets	Google Secret Manager, mounted into the Cloud Run revision

[!IMPORTANT] There is no Firebase anywhere in this product — no Hosting, no Auth, no Functions, no Firestore, no App Check, no Blaze/Spark plan. Auth, the datastore, the API and hosting are all first-party or plain GCP. Any doc that still says otherwise is stale; fix it against this page. Payments are **Moyasar**, never Stripe.

Domains

Domain	Status	Notes
<code>flygaca.com</code>	Owned, canonical	The main product. Marketing domains consolidate here.
<code>api.flygaca.com</code>	The API origin	The Cloud Run service. Mirrors proxy <code>/api/*</code> here.

Domain	Status	Notes
<code>captadel.com</code>	Owned (secured 2026-05-23)	Captain Adel's marketing / standalone front door; redirects to <code>flygaca.com</code> for traffic still hitting the Vercel mirror. Registrar not yet recorded — write it in here when known.
<code>flygaca.sa</code>	Unblocked (entity exists) — see <code>flygaca-sa-registration-readiness.md</code>	Register after the P0-2 name opinion; the "gaca" substring may trigger SaudiNIC name review. Redirect-only → <code>flygaca.com</code> .

PDPL data boundary — the load-bearing rule

All personal data — accounts, profiles, the logbook, **real user queries** — stays in the Kingdom (Cloud SQL + Cloud Run, `me-central2`). Public corpus work can happen on the EU VPS. — `phase0.md`, P0-6, restated for the current stack

That makes the choice unambiguous:

What	Where	Why
Cloud SQL (Postgres)	<code>me-central2</code> (Dammam)	Holds every account, profile, logbook and entitlement row.
Cloud Run (the Fly GACA API + Captain Adel gateway)	<code>me-central2</code> (Dammam)	Receives personal data (user queries).
Cloud Storage (SPA bucket + corpus bucket)	<code>me-central2</code> (Dammam)	Public artifacts, but kept co-regional for latency and simplicity.
Public corpus pipeline / RAG chunk build / evals / staging	EU VPS (Paris) — Hostinger	Public data only ; explicitly never personal data.
ALLaM GPU endpoint (planned)	KSA / in-Kingdom GPU host	Same reason as the gateway — see <code>ay2m/Captain-Adel</code> <code>deploy/allam-vllm.md</code> .

[!WARNING] `me-central1` is Doha, Qatar — it is **NOT in-Kingdom**. Never describe it as Riyadh, Saudi or "the Kingdom box". Any earlier doc wording that pairs `me-central1` with a PDPL claim is wrong on its face. `me-central2` (Dammam) is the canonical answer.

Cloud Run and Cloud SQL must stay in the **same** region: the service reaches the database over a Cloud SQL unix socket (`/cloudsql/PROJECT:REGION:INSTANCE`), so there is no IP allowlist and no proxy sidecar. There is no region constant in the codebase to keep in sync — both are regional resources fixed at deploy time.

Google Cloud

Item	Value
Project	Set at deploy time; the runbook provisions <code>flygaca</code> (the old <code>flygaca-firebase</code> / <code>flygaca-com</code> / <code>fly-gaca-495116</code> projects are dead — do not cite them)
Region	<code>me-central2</code> (Dammam) for Cloud Run, Cloud SQL and the buckets
Cloud Run service	<code>flygaca-api</code> — built from <code>server/</code> at the repo root
Cloud SQL instance	<code>flygaca-db</code> (POSTGRES_16); database <code>flygaca</code> ; schema in <code>server/migrations/</code>
SPA bucket	<code>flygaca-web</code> behind an HTTPS load balancer (bucket backend on <code>/*</code> , serverless NEG on <code>/api/*</code>)
Corpus bucket	The regulatory JSON corpus is offloaded to a bucket and served network-first
Secrets	Secret Manager: <code>session-secret</code> , <code>database-url</code> , <code>google-oauth-secret</code> , <code>genai-api-key</code> , <code>moyasar-secret-key</code> , <code>moyasar-webhook</code> , <code>mail-api-key</code> , <code>cron-secret</code>
Scheduler	<code>flygaca-renewals</code> — daily <code>POST /api/billing/renew</code> with <code>X-Cron-Secret</code>
Config surface	Plain env vars on the Cloud Run revision (<code>server/src/config.ts</code> is the only place the server reads <code>process.env</code>)

Full provisioning sequence: `docs/RUNBOOK-deploy.md` in the product repo, mirrored here as `runbooks/runbook-deploy.md`. Secret and price **names** are mapped in `secrets-and-keys-placement.md`.

Mirrors (all proxy back to the same Cloud Run origin)

A Cloudflare Worker and the Netlify / Vercel mirrors each serve the same `dist/` build and rewrite `/api/*` back to `https://api.flygaca.com` as a same-origin proxy — so chat and account keep working and the strict CSP (`connect-src 'self'`) never changes. Keep any new API surface under `/api/*` for that to hold. The mirrors `X-Robots-Tag: noindex` any host that isn't `flygaca.com`.

EU VPS (Paris) — public-data box

Item	Value
Provider	Hostinger KVM 2 (2 vCPU / 8 GB / 100 GB)
IP	72.62.20.20
Hostname	srv1209075.hstgr.cloud
Region	Paris, Île-de-France (Hostinger, AS47583)

Item	Value
OS	Ubuntu 24.04 LTS, hardened (SSH key-only, ufw, fail2ban)
Role	Corpus ingest / RAG chunk build / eval harness / staging — public data only

GitHub

All product code is under the `ay2m` account. There is no `FLyGACA` org — those paths are legacy redirects.

Repo		What it is
<code>ay2m/FLyGACA</code>	private	The product. The bilingual web app (React 19 + Vite) <i>and</i> its Express backend on Cloud Run. The regulatory corpus and content pipelines live here too.
<code>ay2m/Captain-Adel</code>	private	The AI flight instructor service behind captadel.com
<code>ay2m/FLyGACA-ios</code>	public	The native SwiftUI family — one shared package, one App Store app per exam module
<code>ay2m/Office</code>	private	This repo — the company documents tree
<code>ay2m/FLyGACA-app</code>	public, archived	The retired predecessor of <code>ay2m/FLyGACA</code> . Read-only. Do not cite it as current.

[!NOTE] The six per-module App Store metadata repos (`PPL` , `CPL` , `IR` , `ATPL` , `ELPT` , `AIP`) that older docs reference **do not exist**. Per-app metadata lives inside `ay2m/FLyGACA-ios`.

Sources

- `FLyGACA/docs/RUNBOOK-deploy.md` — the authoritative provisioning + deploy sequence
- `FLyGACA/docs/DATA-HOSTING.md` — how the corpus bucket is served
- `FLyGACA/server/migrations/0001_init.sql` — the schema, with each table's Firestore-era ancestor noted
- `00-strategy/phase0.md` — P0-2 (domain decisions), P0-6 (VPS + PDPL boundary)
- `flygaca-resume-briefing-2026-05-23.md` — hosting + PDPL summary (pre-dates the Cloud Run port; regions still hold, the Firebase references do not)